Ex no: 5(c)	Creating a Custom Docker Image and Uploading It to Docker Hub
Date:	

Aim:

To create a custom Docker image from a simple application, test it locally, and upload (push) the image to Docker Hub for sharing and deployment.

Requirements / Tools

- PC with Docker installed

  Docker Desktop (Windows/Mac)

  Docker Engine (Linux)
- Internet connection
- Docker Hub account
- Text editor

  Notepad

  Visual Studio Code (VS Code)
- Terminal

  PowerShel

  Command Prompt

  Bash
- A simple application (Python or Node.js)

Option 1: Python Application

myapp/

├── app.py

├── Dockerfile

└── .dockerignore

app.py

print("Hello Docker Hub!!!")

Dockerfile

FROM python:3.12-slim

WORKDIR /app

COPY . .

CMD ["python", "app.py"]

.dockerignore

__pycache__/

*.pyc

*.log

Option 2: Node.js Application

myapp-node/

├── package.json

├── index.js

├── Dockerfile

└── .dockerignore

package.json

{

  "name": "myapp-node",

  "version": "1.0.0",

  "main": "index.js"}

index.js

console.log("Hello from Docker!");

Dockerfile

FROM node:18-slim

WORKDIR /app

COPY . .

CMD ["node", "index.js"]

.dockerignore

node_modules

npm-debug.log

Procedure

Step 1: Verify Docker Installation

Open PowerShell, Command Prompt, or Bash and run:

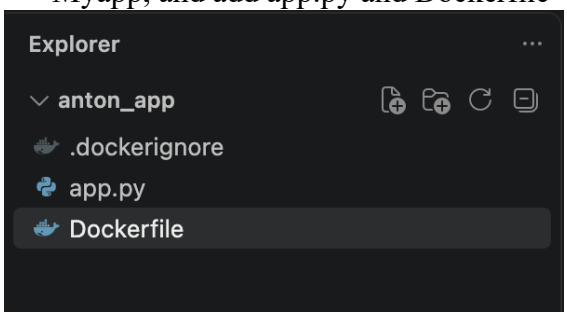
docker –version

```
nevesaaxd@Nevesas-MacBook-Air ~ % docker --version
Docker version 29.7.2, build a7dcaa6
nevesaaxd@Nevesas-MacBook-Air ~ %
```

Step 2: Create the Project Folder and Files

Create a folder named:

Myapp, and add app.py and Dockerfile



Step 3: Build the Docker Image Locally

Navigate to the project folder:

Build the image:

docker build -t anton_app .

```
nevesaaxd@Nevesas-MacBook-Air anton_app % docker build -t anton_app .
[+] Building 14.2s (9/9) FINISHED                                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                             0.0s
 => => transferring dockerfile: 108B                                                             0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                              4.3s
 => [auth] library/python:pull token for registry-1.docker.io                                    0.0s
 => [internal] load .dockerignore                                                                0.0s
 => => transferring context: 64B                                                                 0.0s
 => [1/3] FROM docker.io/library/python:3.12-slim@sha256:78387bc3881b8273120a12ebe6c1ab22b018ccc2c9adf565  9.7s
 => => resolve docker.io/library/python:3.12-slim@sha256:78387bc3881b8273120a12ebe6c1ab22b018ccc2c9adf565  0.0s
 => => sha256:8aff2d3a9af8ed70ae2aa065663f6a7b99d3cd41564528e8d5f039ec0faae595 12.05MB / 12.05MB  3.2s
 => => sha256:3550292b2150d689c8a69849320a815d1d54883f6aecf670f098c1a16f2c5436 251B / 251B        0.5s
 => => sha256:ab2cb3ee67af16dd0212f1564a3c1a7bfc96172b7c21c74895492bd87456fb24 4.61MB / 4.61MB    4.5s
 => => sha256:bf7af0229701decd1b9f42143504fc8f69e5664c37e57001d198e731e4f86c2e 30.16MB / 30.16MB  9.2s
 => => extracting sha256:bf7af0229701decd1b9f42143504fc8f69e5664c37e57001d198e731e4f86c2e         0.3s
 => => extracting sha256:ab2cb3ee67af16dd0212f1564a3c1a7bfc96172b7c21c74895492bd87456fb24         0.0s
 => => extracting sha256:8aff2d3a9af8ed70ae2aa065663f6a7b99d3cd41564528e8d5f039ec0faae595         0.1s
 => => extracting sha256:3550292b2150d689c8a69849320a815d1d54883f6aecf670f098c1a16f2c5436         0.0s
```

Step 4: Verify the Image List available Docker images

docker images

```
nevesaaxd@Nevesas-MacBook-Air anton_app % docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest        28bd5fe8b56d       13.6MB         4.27MB    U
anton_app:latest     0f5d025d23db        214MB         46.8MB
hello-world:latest   5dd0d3e6e255       22.6kB         10.3kB    U
mongo:7              b6421fd6d1c5       1.13GB          286MB    U
myimage:latest       58fe67825a98       13.5MB         4.19MB    U
nevesaaxd@Nevesas-MacBook-Air anton_app %
```

Step 5: Run the Image Locally Command:

docker run --rm my-simple-image1

```
nevesaaxd@Nevesas-MacBook-Air anton_app % docker run --rm anton_app
Hello Docker Hub!!!
nevesaaxd@Nevesas-MacBook-Air anton_app %
```

Step 6: Create a Repository on Docker Hub

1. Open Docker Hub.
2. Sign in to your account.
3. Click **Create Repository**.

4. Enter repository name:

my-simple-image

5. Select:

Public

6. Click **Create**.

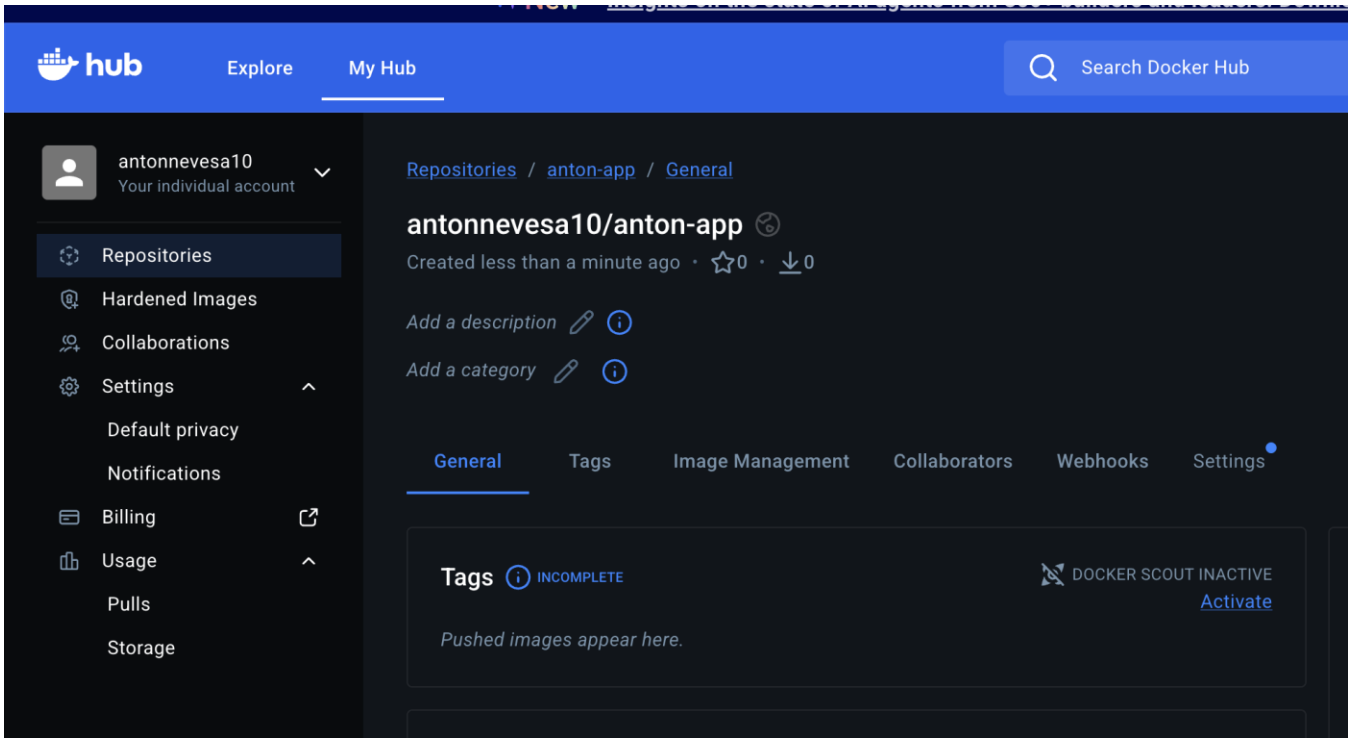
Step 7: Log in to Docker Hub

Open the terminal and run:

docker login

Enter your:

- Docker Hub Username
- Docker Hub Password



Step 8: Tag the Docker Image

If the image was built as myapp:1.0, tag it with your Docker Hub username:

docker tag my-simple-image1 <dockerhub-username>/my-simple-image1

```
nevesaaxd@Nevesas-MacBook-Air anton_app % docker login
Authenticating with existing credentials... [Username: antonnevesa10]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
nevesaaxd@Nevesas-MacBook-Air anton_app % docker tag anton_app antonnevesa10/anton_app
```

Step 9: Push the Image to Docker Hub

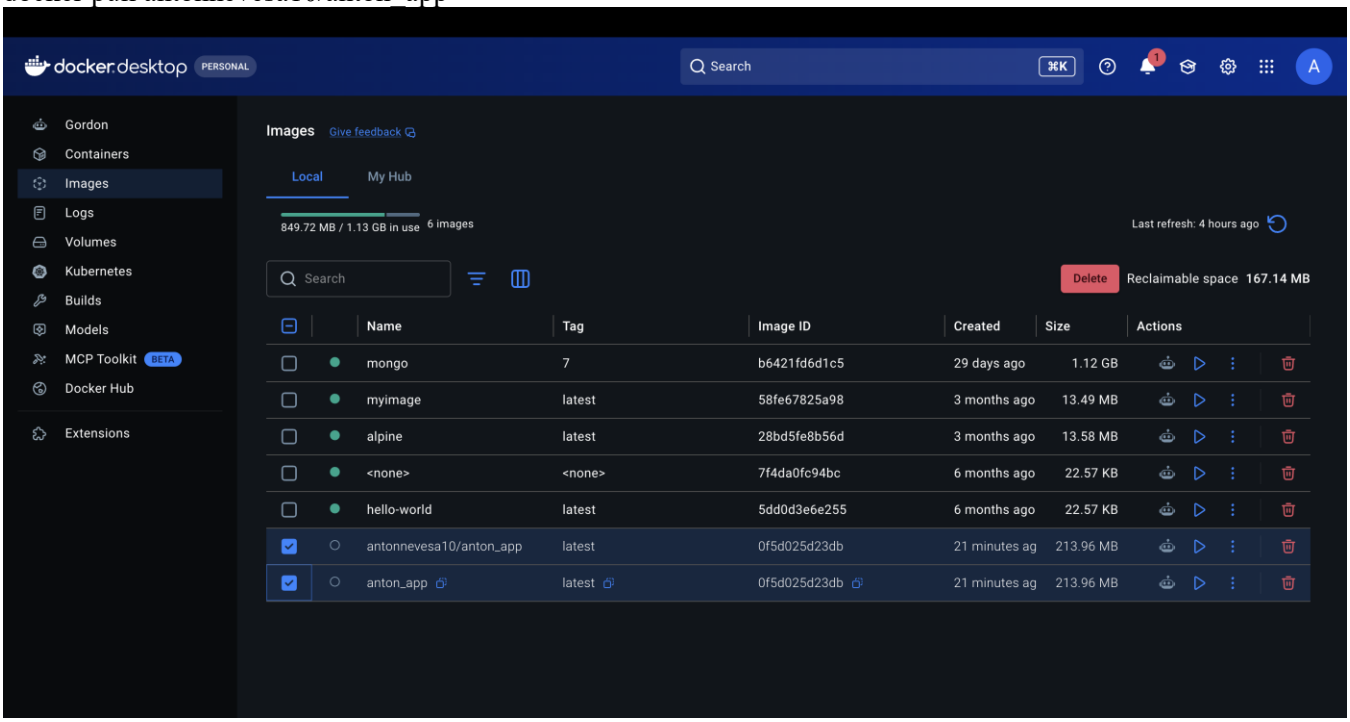
Upload the image:

docker push <dockerhub-username>/my-simple-image1

Step 10: Verify the Image from Docker Hub

Pull the image from Docker Hub:

docker pull antonnevesa10/anton_app



```
nevesaaxd@Nevesas-MacBook-Air anton_app % docker pull antonnevesa10/anton_app
Using default tag: latest
latest: Pulling from antonnevesa10/anton_app
Digest: sha256:0f5d025d23db4b508b7c2a53879942dd991ac94ba522eaf70f8f5ba76878fa15
Status: Image is up to date for antonnevesa10/anton_app:latest
docker.io/antonnevesa10/anton_app:latest
nevesaaxd@Nevesas-MacBook-Air anton_app %
```

Rubrics:

Problem Understanding & approach (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)

Result:

A custom Docker image was successfully created from a simple application, tested locally, tagged, uploaded to Docker Hub, and verified by pulling and running it from the repository